

# Software Requirement Specification (SRS)

## AI-Multi Language Translator Web Application

### 1. Introduction

The purpose of this project is to develop a web-based AI language translation application that supports translation between English and multiple languages using Natural Language Processing (NLP) models.

The system uses pretrained Helsinki-NLP transformer models available through the Hugging Face Transformers library.

The application allows users to enter text, select source and target languages, and receive translated output through a simple web interface with login and signup functionality.

### 1.2 Scope

The system will accept English text input, support translation across English, Hindi, French, Spanish, and German and display translated text on the webpage, use a pre-trained NLP model, and provide a simple user-friendly web interface. The system will be deployed online using Hugging Face Spaces cloud hosting platform.

### 1.3 Definitions

| Term              | Definition                            |
|-------------------|---------------------------------------|
| NLP               | Natural Language Processing           |
| SRS               | Software Requirement Specification    |
| API               | Application Programming Interface     |
| UI                | User Interface                        |
| Flask             | Python web framework                  |
| Transformer Model | Deep learning model used in NLP tasks |

## 1.4 Technologies Used

| Technology                | Purpose                      |
|---------------------------|------------------------------|
| Python                    | Backend programming language |
| Flask                     | Web framework                |
| Hugging Face Transformers | NLP model integration        |
| HTML                      | Webpage structure            |
| CSS                       | User interface styling       |
| Hugging Face Spaces       | Cloud deployment platform    |
| JSON                      | User data storage            |

## 2. Overall Description

The system is a standalone web application where users interact through a browser interface. The frontend sends user input and selected language preferences to the Flask backend server, which processes translation using Hugging Face NLP transformer models. The translated output is then returned and displayed on the webpage. The system also includes user authentication with login and signup functionality for secure access to the translation interface.

### 2.2 Product Functions

- User signup and login
- Accept multilingual text input
- Select source and target language
- Process translation using NLP models
- Display translated text and copy option
- Allow multiple translations

### 3. System Features

#### Feature 1: Text Translation

- Input: Text entered by the user along with selected source and target languages.
- Process: Text is processed using Hugging Face NLP transformer translation models.
- Output: Translated text displayed on the webpage.

#### Feature 2: User Authentication

- Input: User email and password.
- Process: System validates user credentials using hashed password storage.
- Output: Secure access to the translation interface.

### 3.2 Functional Requirements

| ID  | Requirement                                 |
|-----|---------------------------------------------|
| FR1 | System shall allow user signup              |
| FR2 | System shall allow user login               |
| FR3 | System shall accept multilingual text input |
| FR4 | System shall translate using NLP models     |
| FR5 | System shall display translated output      |
| FR6 | System shall allow multiple translations    |

### 4. External Interface Requirements

User Interface components include:

- Signup Page
- Login Page
- Language dropdown selection
- Textbox for entering selected language's text
- Translate button to trigger translation
- Output box to display translated text with a copy icon to copy translated text
- Logout button

## 5. Non-Functional Requirements

- Performance: Translation should occur within few seconds depending on model inference.
- Security: No personal data stored and input validation should be applied.
- Usability: Interface should be simple and easy to understand.
- Reliability: System should handle translation requests without crashes.

## 6. Deployment

The application will be deployed using Hugging Face Spaces.

Steps:

1. Push project to GitHub
2. Connect repository to Render
3. Configure Python environment
4. Deploy Flask application
5. Access application via public URL

## 7. Future Enhancements

- Support multiple regional languages
- Add speech-to-text translation input
- Improve UI design
- Maintain translation history
- Improved translation accuracy

## 8. Conclusion

The AI Multi-Language Translator Web Application provides a simple and efficient solution for language translation using Natural Language Processing (NLP).

By integrating Flask, Hugging Face Transformers, and Helsinki NLP translation models, the system enables translation across multiple languages through an interactive web interface. The application also includes user authentication features and cloud deployment using Hugging Face Spaces, making the system accessible, user-friendly, and scalable.